

last time: complex numbers, many examples of vector spaces

today: norms, inner products, Cauchy-Schwarz inequality, orthogonality, orthonormal basis, Gram-Schmidt procedure

norms: A norm on a vector space V is a mapping $\|\cdot\|: V \rightarrow [0, \infty)$ s.t.

$$\|u+v\| \leq \|u\| + \|v\| \quad \text{triangle inequality}$$

$$\|\alpha u\| = |\alpha| \|u\| \quad \text{homogeneity}$$

$$\|u\| = 0 \Leftrightarrow u = 0 \quad \text{positive definiteness}$$

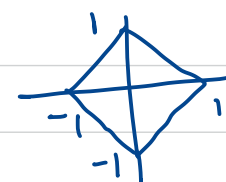
note that $\|u\|$ is real and non-negative even if V is a complex vector space

The distance from u to v is simply $\|u-v\|$

norms on $\mathbb{R}^n$ or $\mathbb{C}^n$:

one-norm
(Manhattan)

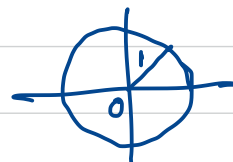
$$\|x\|_1 = \sum_{k=1}^n |x_k|$$



2-norm
(Euclidean)

$$\|x\|_2 = \sqrt{\sum_{k=1}^n |x_k|^2}$$

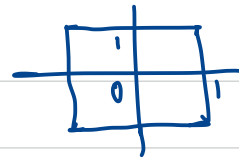
↑
Pythagorean theorem



shape of unit "ball"

∞ -norm
(max)

$$\|x\|_{\infty} = \max_{1 \leq k \leq n} |x_k|$$



The unit ball in V is defined as

$$B = \{x \in V : \|x\| \leq 1\}$$

claim: for any $x \in \mathbb{C}^n$, $\|x\|_{\infty} \leq \|x\|_2 \leq \|x\|_1$

proof: $\|x\|_{\infty} = \sqrt{\max_k |x_k|^2} \leq \sqrt{\sum_k |x_k|^2} = \|x\|_2$

and $\|x\|_2 = \|x_1 e_1 + \dots + x_n e_n\|_2$
 $\leq \|x_1 e_1\|_2 + \dots + \|x_n e_n\|_2$
 $= |x_1| \underbrace{\|e_1\|_2}_1 + \dots + |x_n| \underbrace{\|e_n\|_2}_1 = \sum_k |x_k| = \|x\|_1$

$(e_k)_j = \delta_{jk} = \begin{cases} 1 & j=k \\ 0 & \text{o.w.} \end{cases}$
elementary unit vectors

e.g. in $\mathbb{R}^2$: $x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = x_1 \underbrace{\begin{pmatrix} 1 \\ 0 \end{pmatrix}}_{e_1} + x_2 \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}_{e_2}$

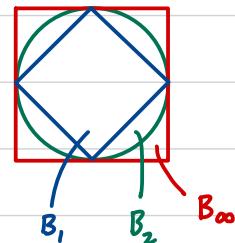
The unit balls are then nested in the opposite direction:

$$B_1 \subseteq B_2 \subseteq B_{\infty}$$

e.g. $x \in B_1 \Leftrightarrow \|x\|_1 \leq 1$

$$\Rightarrow \|x\|_2 \leq \|x\|_1 \leq 1$$

$$\Rightarrow x \in B_2$$



So a smaller unit ball means the norm is larger.

Here are the norms of some of the other spaces we discussed:

$$\ell^1(\mathbb{N}_0): \quad \|x\| = \sum_{n=0}^{\infty} |x_n|$$

$$\ell^2(\mathbb{N}): \quad \|x\| = \sqrt{\sum_{n=1}^{\infty} |x_n|^2}$$

$$\ell^1[a, b]: \quad \|f\| = \int_a^b |f(x)| dx$$

$$\ell^2[a, b]: \quad \|f\| = \sqrt{\int_a^b |f(x)|^2 dx}$$

$$\ell^\infty[a, b]: \quad \|f\| = \max_{a \leq x \leq b} |f(x)|$$

The 2-norms are special because they come from an inner product

$$\left. \begin{array}{l} \mathbb{C}^n: \quad \langle x, y \rangle = \sum_{k=1}^n x_k \overline{y_k} \\ \ell^2(\mathbb{Z}): \quad \langle x, y \rangle = \sum_{k=-\infty}^{\infty} x_k \overline{y_k} \end{array} \right\} \quad \|x\| = \sqrt{\langle x, x \rangle}$$

$$\ell^2(\mathbb{R}): \quad \langle f, g \rangle = \int_{-\infty}^{\infty} f(x) \overline{g(x)} dx, \quad \|f\| = \sqrt{\langle f, f \rangle}$$

An inner product is a mapping $\langle \cdot, \cdot \rangle: V \times V \rightarrow \mathbb{C}$ such that

1. if $u \neq 0$ then $\langle u, u \rangle > 0$ positivity

(part of the requirement here is that $\langle u, u \rangle$ is real)

$$2. \langle v, u \rangle = \overline{\langle u, v \rangle}$$

conjugate symmetry

$$3. \langle \alpha u, v \rangle = \alpha \langle u, v \rangle$$

homogeneity

$$4. \langle u+v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$$

linearity with respect to the first slot

additional properties:

$$(5) \langle 0, v \rangle = 0 \quad (\text{follows from (3)})$$

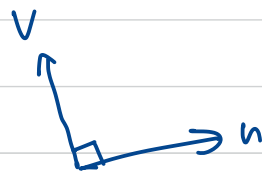
$$\left. \begin{array}{l} (6) \langle u, \alpha v \rangle = \overline{\alpha} \langle u, v \rangle \quad (\text{follows from (2\&3)}) \\ (7) \langle w, u+v \rangle = \langle w, u \rangle + \langle w, v \rangle \quad (u \text{ " (2\&4)}) \end{array} \right\} \text{conjugate linearity w.r.t. 2nd slot}$$

For a real vector space, drop the complex conjugate in 2, 6

real case: $\langle \cdot, \cdot \rangle$ is bilinear

complex case: $\langle \cdot, \cdot \rangle$ is sesquilinear

def: if $\langle u, v \rangle = 0$, u and v are orthogonal



note: over $\mathbb{C}$, it's important to conjugate the second

slot. $\langle x, y \rangle = x^T \bar{y}$ ($\begin{pmatrix} 1 \\ i \end{pmatrix}$ is not orthogonal to itself)

In Physics, one usually defines inner products to be

linear in the second slot and conjugate linear in the first

(just a convention)

$$\text{e.g. } \langle x, y \rangle_{\text{physics}} = \bar{x}^T y$$

The norm on an inner product space is given by

$$\|u\| = \sqrt{\langle u, u \rangle} \quad \text{from } \textcircled{*}$$

homogeneity: $\|\alpha u\| = \sqrt{\langle \alpha u, \alpha u \rangle} = \sqrt{\alpha \bar{\alpha} \langle u, u \rangle}$
 $= \sqrt{|\alpha|^2 \langle u, u \rangle} = |\alpha| \sqrt{\langle u, u \rangle} = |\alpha| \cdot \|u\|$

positive definiteness: $u = 0 \xRightarrow{\text{by (5)}} \langle u, u \rangle = 0 \xRightarrow{\text{from } \textcircled{*}} \|u\| = 0$
 $\|u\| = 0 \xRightarrow{\text{from } \textcircled{*}} \langle u, u \rangle = 0 \xRightarrow{\text{by (1)}} u = 0$

to prove the triangle inequality, we first need to prove the Cauchy-Schwarz inequality:

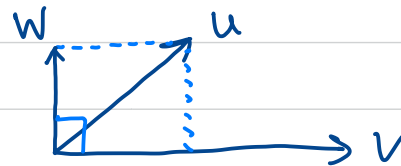
$$\text{Cauchy-Schwarz: } |\langle u, v \rangle| \leq \|u\| \cdot \|v\|$$

geometric proof (see book for calculus-based proof)

The theorem is trivial if $u=0$ or $v=0$. ($0 \leq 0$ ✓)

so assume $u \neq 0$ and $v \neq 0$.

$$\text{Let } w = u - \frac{\langle u, v \rangle}{\langle v, v \rangle} v$$



Then

$$0 \leq \langle w, w \rangle = \left\langle u - \frac{\langle u, v \rangle}{\langle v, v \rangle} v, u - \frac{\langle u, v \rangle}{\langle v, v \rangle} v \right\rangle$$

$$\begin{aligned}
& \stackrel{\text{foil}}{=} \langle u, u \rangle - \frac{\overline{\langle u, v \rangle}}{\langle v, v \rangle} \langle u, v \rangle - \frac{\langle u, v \rangle}{\langle v, v \rangle} \underbrace{\langle v, u \rangle}_{\overline{\langle u, v \rangle}} + \frac{\langle u, v \rangle \overline{\langle u, v \rangle}}{\langle v, v \rangle^2} \langle v, v \rangle \\
& = \langle u, u \rangle - \frac{|\langle u, v \rangle|^2}{\langle v, v \rangle} \quad \therefore |\langle u, v \rangle|^2 \leq \langle u, u \rangle \langle v, v \rangle
\end{aligned}$$

alternative proof: let $\alpha = \frac{\langle u, v \rangle}{\langle v, v \rangle}$ so $w = u - \alpha v$

$$\langle w, v \rangle = \langle u, v \rangle - \alpha \langle v, v \rangle = 0$$

$$\begin{aligned}
0 \leq \langle w, w \rangle &= \langle w, u - \alpha v \rangle = \langle w, u \rangle - \underbrace{\alpha \langle w, v \rangle}_0 \\
&= \langle u - \alpha v, u \rangle = \langle u, u \rangle - \alpha \underbrace{\langle v, u \rangle}_{\overline{\langle u, v \rangle}} \\
&= \|u\|^2 - \frac{|\langle u, v \rangle|^2}{\|v\|^2}
\end{aligned}$$

Note that the equality holds iff $w = 0$, so that

$$u = \frac{\langle u, v \rangle}{\langle v, v \rangle} v, \text{ i.e., iff } u \text{ and } v \text{ are colinear} \\
(\exists \alpha \in K \text{ s.t. } u = \alpha v)$$

Geometric interpretation (over $\mathbb{R}$)

$$\langle u, v \rangle = \|u\| \cdot \|v\| \cdot \underbrace{\cos \theta}_{\text{between } -1 \text{ and } 1}$$



Consequence of Cauchy-Schwarz:

the triangle inequality holds: $\|u+v\| \leq \|u\| + \|v\|$

so $\|u\| = \sqrt{\langle u, u \rangle}$ is a norm.

$$\|u+v\|^2 = \underbrace{\langle u, u \rangle}_{\|u\|^2} + \underbrace{\langle u, v \rangle + \langle v, u \rangle}_{\textcircled{*}} + \underbrace{\langle v, v \rangle}_{\|v\|^2}$$

let $c = \langle u, v \rangle$. Then $\langle u, v \rangle + \langle v, u \rangle = c + \bar{c} = 2\operatorname{Re} c$

$$(c = a+ib, \bar{c} = a-ib, c+\bar{c} = 2a)$$

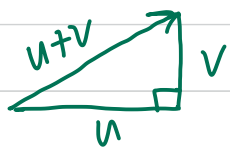
$$\therefore \textcircled{*} \text{ is real and } \textcircled{*} = 2a \leq 2|a| = 2\sqrt{a^2} \leq 2\sqrt{a^2+b^2} \\ = 2|c| = 2|\langle u, v \rangle|$$

$$\therefore \|u+v\|^2 \leq \|u\|^2 + 2|\langle u, v \rangle| + \|v\|^2$$

$$\stackrel{\text{C-S}}{\leq} \|u\|^2 + 2\|u\| \cdot \|v\| + \|v\|^2 \leq (\|u\| + \|v\|)^2$$

property of the norm $\|u\| = \sqrt{\langle u, u \rangle}$:

Pythagorean theorem: If $\langle u, v \rangle = 0$ then $\|u+v\|^2 = \|u\|^2 + \|v\|^2$



$$\langle u+v, u+v \rangle = \underbrace{\langle u, u \rangle}_0 + \underbrace{\langle u, v \rangle}_0 + \underbrace{\langle v, u \rangle}_0 + \langle v, v \rangle$$

Def: vectors $e_1, \dots, e_N$ are orthogonal if $\langle e_i, e_j \rangle = 0$ if $i \neq j$

$$\text{orthonormal if } \langle e_i, e_j \rangle = \delta_{ij} = \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases}$$



Kronecker delta
(discrete delta function)

a basis for a vector space
or a subspace is a linearly
independent spanning set.

A nice feature of an orthonormal basis is that you can compute expansion coefficients by taking inner products:

Suppose $e_1, \dots, e_m$ is an O.N. basis for V and $x \in V$.

$$\text{write } x = \sum_{j=1}^m \alpha_j e_j$$

$$\text{Then } \langle x, e_k \rangle = \left\langle \sum_j \alpha_j e_j, e_k \right\rangle = \sum_j \alpha_j \underbrace{\langle e_j, e_k \rangle}_{\delta_{jk}} = \alpha_k$$

formula: $\alpha_k = \langle x, e_k \rangle$

Gram-Schmidt orthogonalization

given $u_1, \dots, u_N$ linearly independent

find $e_1, \dots, e_N$ orthonormal with the same span

procedure: $e_1 = u_1 / \|u_1\|$

$$E_2 = u_2 - \langle u_2, e_1 \rangle e_1, \quad e_2 = \frac{E_2}{\|E_2\|}$$

$$E_3 = u_3 - [\langle u_3, e_1 \rangle e_1 + \langle u_3, e_2 \rangle e_2], \quad e_3 = \frac{E_3}{\|E_3\|}$$

etc.

Justification: let $S(n)$ be the statement

" $e_1, \dots, e_n$ is an O.N. basis for $\text{span}(u_1, \dots, u_n)$ "

We want to prove that $S(N)$ is true.

base case: ($n=1$) $e_1 = \frac{u_1}{\|u_1\|}$

it's a unit vector and a basis for the one-dimensional subspace spanned by u_1 . $\therefore S(1)$ is true.

(unit vector: $\|\frac{u_1}{\|u_1\|}\| = \frac{1}{\|u_1\|} \|u_1\| = 1$)

induction step.

Suppose $2 \leq n \leq N$ and $S(n-1)$ is true.

E_n is defined by $E_n = u_n - \sum_{j=1}^{n-1} \langle u_n, e_j \rangle e_j$ since $S(n-1)$ is true

E_n is non-zero since otherwise $u_n = \sum_{j=1}^{n-1} \langle u_n, e_j \rangle e_j \in \text{span}(u_1, \dots, u_{n-1})$

this contradicts $u_1, \dots, u_N$ being linearly independent

let $k \in \{1, \dots, n-1\}$. Then

$$\begin{aligned} \langle E_n, e_k \rangle &= \langle u_n, e_k \rangle - \left\langle \sum_{j=1}^{n-1} \langle u_n, e_j \rangle e_j, e_k \right\rangle \\ &= \langle u_n, e_k \rangle - \sum_{j=1}^{n-1} \langle u_n, e_j \rangle \underbrace{\langle e_j, e_k \rangle}_{\delta_{jk}} \\ &= \langle u_n, e_k \rangle - \langle u_n, e_k \rangle = 0 \end{aligned}$$

$e_n = \frac{E_n}{\|E_n\|}$ satisfies $\|e_n\|=1$ and $\langle e_n, e_k \rangle = 0$ for $1 \leq k \leq n-1$

$\therefore e_1, \dots, e_n$ are orthonormal and span the same space as $u_1, \dots, u_n$

$\therefore S(n)$ is true. $S(1) \Rightarrow S(2) \Rightarrow S(3) \Rightarrow \dots \Rightarrow S(N)$

so $S(N)$ is true as claimed.